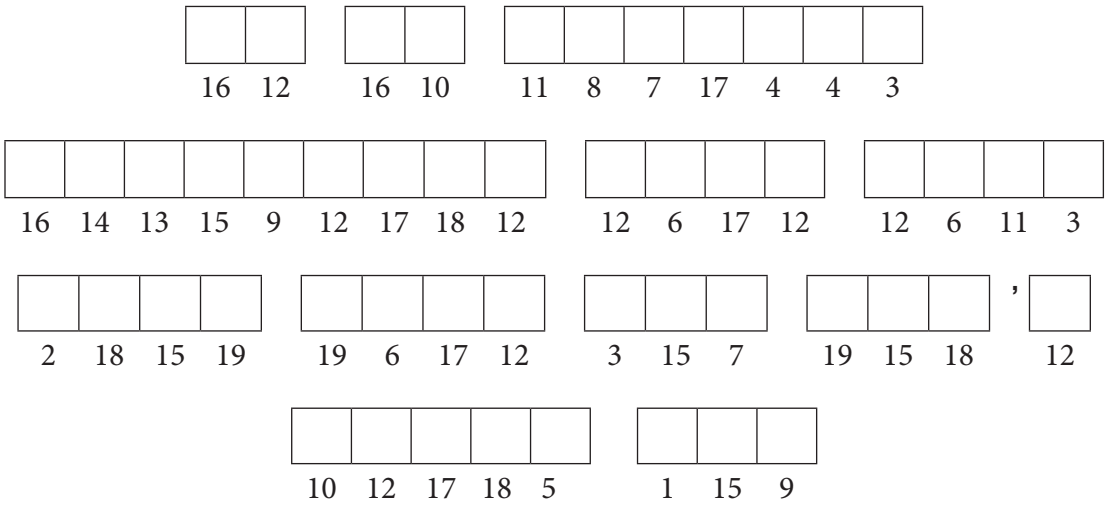
WORKSHEET

Chapter 3: Composition of functions

It is important that people know what you stand for …



Mary H. Waldrip

Questions

Given $f(x) = 3x - 7$, $g(x) = 2x^2 + 5$, $h(x) = 4x + 1$, $i(x) = x + 10$ and $j(x) = (5x - 2)^2$, find:

1 $fg(x)$

2 $f \circ h(x)$

3 $f[i(x)]$

4 $hj(x)$

© Cengage Learning Australia Pty Ltd 2014

5 $g \circ f(x)$

6 $i \circ f(x)$

7 $ji(x)$

8 $i[h(x)]$

9 $f^{-1}(x)$

10 $f \circ f^{-1}(x)$

11 $f\,[f^{-1}(5)]$

12 $g^{-1}(x)$

13 $g^{-1}[g(-7)]$

14 $h \circ f(x)$

15 $i^{-1}(x)$

16 $i^{-1}g(x)$

17 $i^{-1}g(3)$

18 $h^{-1} \circ f(x)$

19 $h^{-1} \circ f(-2)$

© Cengage Learning Australia Pty Ltd 2014

Solutions

A 13

D $18x^2 - 84x + 103$

E 5

F $6x^2 + 8$

H $3x + 3$

I $2x^2 - 5$

K $12x - 4$

L $100x^2 - 80x + 17$

M $12x - 27$

N $\frac{3x - 8}{4}$

O $x - 10$

P -7

Q $4x + 11$

R $\frac{x + 7}{3}$

S x

T $\sqrt{\frac{x-5}{2}}$

U $25x^2 + 480x + 2304$

W $-3\frac{1}{2}$

Y $3x + 23$

© Cengage Learning Australia Pty Ltd 2014